

Unit 2 Homework — Exponent Rules

Complete independently · Exponent Rules

A. Exponent Rules — Product & Quotient

Problem 1

Simplify: $x^5 \cdot x^6$

$$x^{11}$$

Problem 2

Simplify: $b^9 \div b^4$

$$b^5$$

Problem 3

Simplify: $(4x^2)(2x^5)$

$$8x^7$$

Problem 4

Simplify: $24m^8 \div 6m^3$

$$4m^5$$

Problem 5

Simplify: $c^3 \cdot c^4 \cdot c^2$

$$c^9$$

Problem 6

Simplify: $(7y^4)(2y)$

$$14y^5$$

B. Power, Zero & Negative Exponents

Problem 1

Simplify: $(y^5)^2$

$$y^{10}$$

Problem 2

Simplify: $(3x^2)^2$

$$9x^4$$

Problem 3

Simplify: 12^0

$$1$$

Problem 4

Simplify: $(8m^3n^2)^0$

$$1$$

Problem 5

Simplify: 2^{-3} (write as a fraction)

$$1/8$$

Problem 6

Simplify: y^{-2} (write as a fraction)

$$\underline{1/y^2}$$

Problem 7

Evaluate -2^4 (no parentheses).

$$\underline{-16}$$